

Shafiya Kausar

AI Engineer · Data Scientist · Machine Learning Engineer

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LinkedIn | GitHub | Portfolio | Medium

Profile

AI Engineer and Data Scientist passionate about transforming complex data into intelligent, real-world solutions. Specialized in NLP, multimodal AI, and predictive modeling, with hands-on experience building scalable, production-ready systems that deliver measurable impact. Known for combining strong analytical thinking with product intuition to design efficient, user-focused AI solutions. Seeking a full-time opportunity to contribute to innovative, high-impact projects in AI and data science.

Skills

Technical: Python, R, SQL, Flask

ML/AI: Deep Learning, NLP, Time-Series, Multimodal AI, Document Intelligence

Libraries: Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch

Tools: Power BI, Git, MLflow, Docker, Google Cloud, Azure

Languages: English (Fluent), French (A1)

Professional Experience

- AI Engineer — INSEAD, Fontainebleau**  12/2025 – Present
- Built an agentic RAG knowledge base unifying administration, rules, code, payments, and HR documentation.
 - Developed an AI agent automating annual resource planning, replacing a manual Google Sheets workflow.
 - Created an AI agent scheduling the on-site AV/IT support team via Outlook and ServiceNow.
- Research Assistant — Aivancity School for Technology, Business & Society, Paris**  01/2025 – 01/2026
- Developed pipelines for extracting text, images, and tables from PDFs.
 - Optimized multi-zone document intelligence systems for structured data extraction.
- Data Scientist — IntHealth, Paris** 05/2025 – 01/2026
- Built mood-to-genre NLP pipeline, increasing user engagement by 35%.
 - Delivered production-ready MVP integrating sentiment analysis with recommendations.
- Machine Learning Engineer Intern — Codveda**  07/2025 – 09/2025
- Improved model performance by 12% through benchmarking and evaluation optimization.
 - Streamlined evaluation workflows and improved consistency of model metrics.
- Product Manager Intern — Progtion, Bangalore**  06/2025 – 09/2025
- Defined and prioritized SaaS analytics roadmap, reducing delivery time by 20%.
 - Analyzed 1000+ user feedback entries to improve usability and product alignment.
- Data Scientist — AURA, Paris**  01/2025 – 04/2025
- Built CNN/RNN architectures for ECG data, improving seizure detection accuracy by 17%.
 - Automated preprocessing pipelines, increasing data throughput by 25%.
- Research Assistant (AI/NLP) — NeuralTeks, Paris**  09/2024 – 12/2024
- Integrated multimodal retrieval models achieving 0.82 mAP.
 - Improved embeddings via prompt optimization and semantic alignment.
- Assistant Manager Operations / Data Intern — AR Cars, Bangalore** 10/2022 – 11/2022
- Built Excel and Power BI dashboards for costs, logistics, and procurement tracking.
 - Increased sales by 37% through interactive dashboards improving stakeholder insights.

Volunteer Experience

- UN Volunteer — UI/UX Designer, Action Lab for Development, Canada**  01/2026 – 05/2026
- Conducted hands-on UX testing of the MoralityCode ethical AI app, identifying 11 critical bugs and usability issues across the user journey.
 - Designed Before/After solution mockups with implementation specs for the developer team, covering input validation, navigation fixes, and flow improvements.
 - Classified all UI pages as mandatory or optional to establish a clear product structure.

Food Distribution Volunteer — Linkee, Paris

09/2024 – 12/2025

- Assist in collecting and distributing meals to people in need and refugees across Paris, contributing to the fight against food waste and food insecurity.

Food Collection Volunteer — Secours Populaire Français, Paris

02/2024 – 03/2024

- Collect surplus food from supermarkets for redistribution to vulnerable communities and refugees through the national solidarity network.

Projects

Medical AI Dashboard

- Streamlit app for medical imaging: DICOM exploration, chest X-ray classification (ResNet50 vs. ViT, ~0.95 AUC) with Grad-CAM, and 3D organ segmentation via MONAI U-Net.

MIMIC-II Disease Phenotyping

- Unsupervised phenotyping of ICU patients (PCA, UMAP, KMeans) with an interactive Streamlit dashboard and CI pipeline.

FedHealth — Federated Learning

- Privacy-preserving federated learning (PyTorch FedAvg) for heart-disease risk prediction across simulated hospitals, with a tested library and Streamlit UI.

WorldMedQA × BiomedCLIP

- Zero-shot multimodal medical question-answering with Microsoft's BiomedCLIP vision-language model on the 568-question WorldMedQA-V benchmark.

Voice HMI ASR Benchmark

- Benchmarked open-source ASR models (Whisper, Wav2Vec2, Vosk) for voice-controlled robotic pick-and-place across accents and noise, with a custom command-recognition metric.

Resume Skill Extractor

- Flask app using the Mistral API to extract and classify résumé skills (95% precision).

Publications

When LLM Signals Hurt: A Coverage-Density Analysis of LLM-Augmented Reinforcement Learning for Stock Trading

2026

- Sole-authored preprint on SSRN and SocArXiv (OSF) — LLM-augmented reinforcement learning for stock trading.

Agentic AI for Predictive Analytics in Industry 5.0: A Technical Review of Algorithms and Use Cases Across Healthcare, Manufacturing, Education, and Sustainable Engineering

Forthcoming

- Book chapter in *Empowering Autonomy: The Shift from Generative to Agentic AI in Collective Intelligence Systems* (CRC Press, Taylor & Francis), accepted by editors and in production.

Talks & Presentations

IANLP 2026 — Int'l Conference on AI & NLP, Hassan II University of Casablanca, Morocco (Online)

06/2026

- Paper accepted for **oral presentation**: “PhenoPrompt: A Prompt-Based Clinical Phenotyping Framework Using Entity Extraction, Unsupervised Clustering, and Retrieval-Augmented Generation” at the inaugural edition of IANLP 2026.

PyConES 2026 — Barcelona, Spain (In-person)

11/2026

- Accepted talk: “Mitigating Hallucination in Scientific LLM Systems: A Python Engineering Framework.”

Education

MSc AI & Data Science — Aivancity, Paris

2024 – Present

BSc Physics, Chemistry & Mathematics — Bengaluru North University, India

2020 – 2023

Certifications

- AI Product Engineering — UC Berkeley SCET
- Digital Transformation in Supply Chains — PMI
- Creating API Documentation — LinkedIn Learning